

EESTER — Module Specification v4

Proof of Learning Journey · Hackathon Build

Version: 4.0 — Added Mentor UI + Approach C linking flow
Stack: Inherits Smart Parenting stack — agent reads codebase before coding
DB Priority: Reuse Smart Parenting models — only define new entities where needed
Audience: Human teammates + AI coding models (Claude Opus / Sonnet)

[AI READING NOTE]

- Sections marked [AI] contain explicit coding instructions. Follow precisely.
- Do NOT invent schema, routes, or logic not described here.
- Import from `shared.model` first. Only create new models when nothing existing works.
- Priority order for every decision: reuse → extend → create new.
- Linking flow is Approach C: AI suggest → Student context note (optional) → Mentor confirm. - **Read requirements.txt and package.json before writing any code.** Use existing libraries. - If Smart Parenting stack is incompatible with any EESTER feature → use best-fit modern alternative. Do not force compatibility. - **If a Smart Parenting library cannot be reused for a new EESTER feature, use the best available alternative — do not force an incompatible tool.**

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1. Concept

EESTER là “con dấu số” của EEST — một living credential được xây dựng dần dần từ nhiều nguồn đóng góp. Không phải snapshot tĩnh — mà là **proof of process** tích lũy theo thời gian.

Tại sao không fake được

Mỗi artifact đều có: - Timestamp thật được hash vào chain - Contributor xác nhận thật (mentor approve, admin verify) - AI trace kết nối artifact — mối quan hệ nhân quả có evidence cụ thể - Auth scoring 7 chiều dựa trên toàn bộ graph — không thể game từng chiều riêng lẻ

Không thể tạo ra 14 tháng hành trình trong 1 đêm — dù AI có giỏi đến đâu.

Pitch line

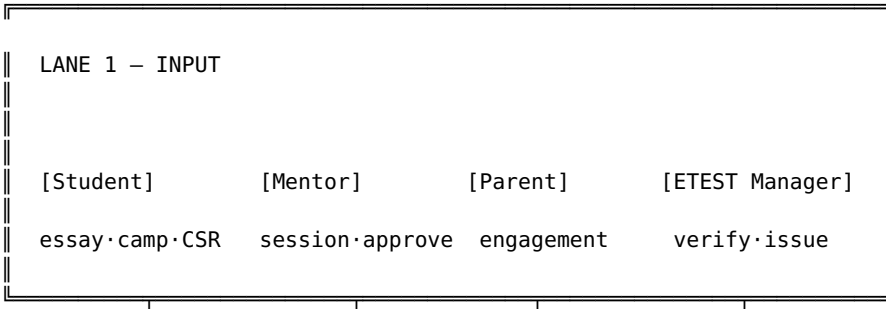
“In a world where AI can write any essay — EESTER is the only proof universities can trust.”

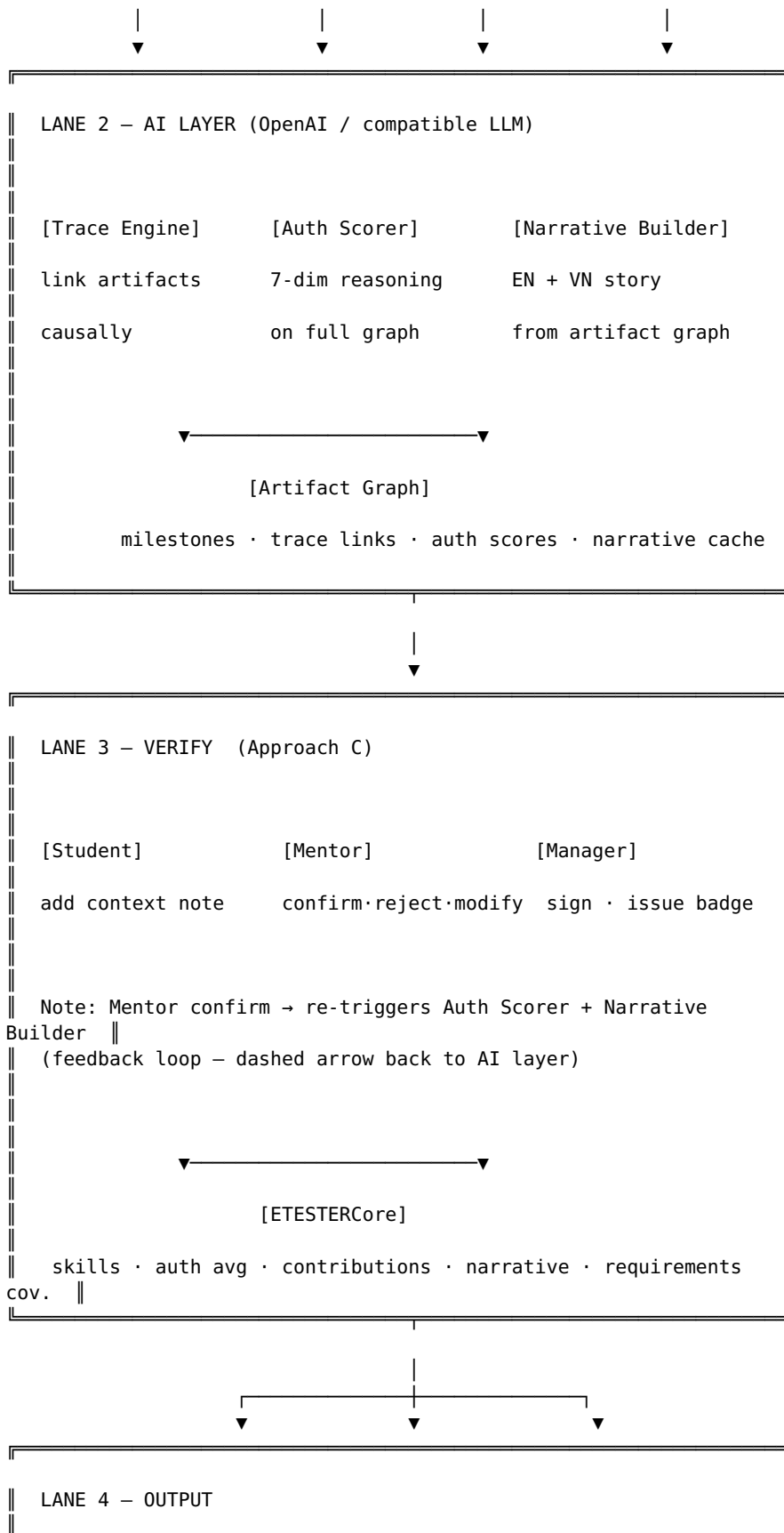
4 Contributors

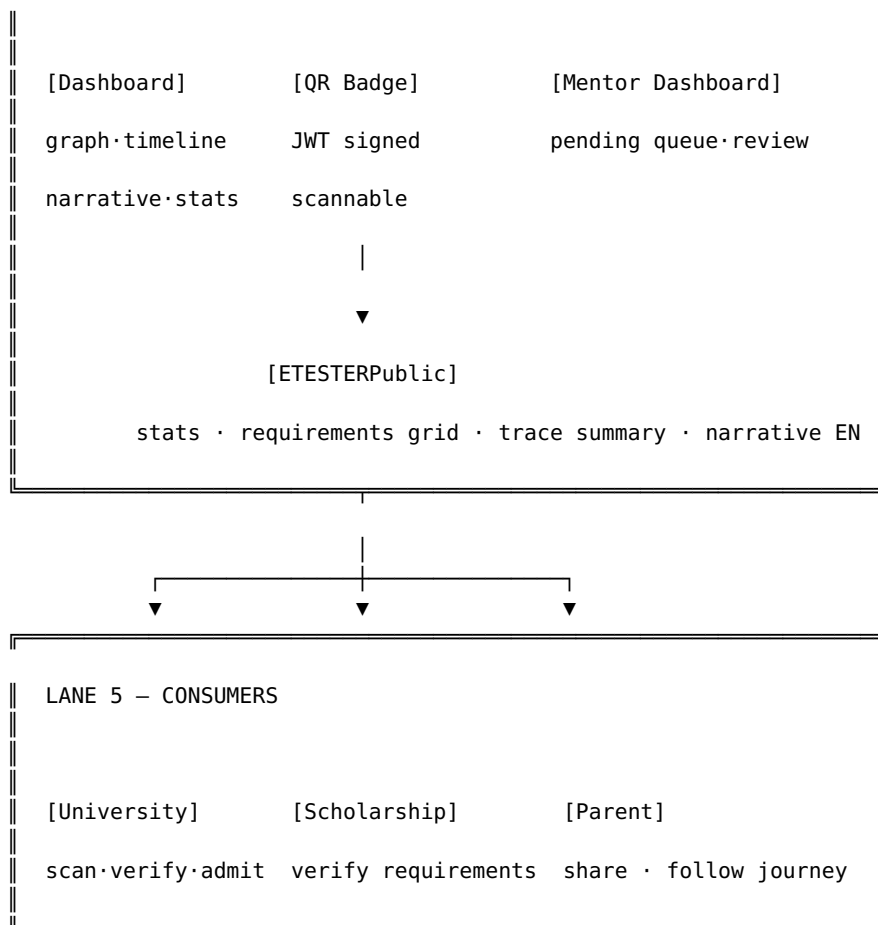
Contributor	Đóng góp	Trust weight
Student	Submit artifacts + context notes cho trace links	Medium
Mentor	Confirm trace links + approve artifacts + session notes	High
Parent	Engagement signal từ Smart Parenting Module 1	Low-Medium
EEST Manager	Đóng dấu chính thức · issue badge · spot-check	Highest

1b. Overall Pipeline

Đây là toàn bộ luồng dữ liệu của EESTER từ input đến consumer.
Click vào mỗi node trong diagram (chat) để xem chi tiết từng phần.







Key:

- Trace Engine → POST /api/etester/trace/:milestoneId (async, internal)
- Auth Scorer → POST /api/etester/score-auth/:milestoneId (after mentor confirm)
- Narrative → POST /api/etester/rebuild-narrative/:studentId (after score-auth)
- Feedback loop → Every mentor confirm re-triggers: Auth → Core → Narrative
- ETESTERCore → Single source of truth, rebuilt on every confirmed event
- QR Badge → JWT RS256 signed payload, verify without account

2. Key Values — Demo highlights

[AI] Đây là những feature **PHẢI** hoạt động trong demo. Không được cut.

#	Feature	Demo moment
KV1	Artifact graph kết nối milestones	Click “Personal Statement FINAL” → thấy trace chain với evidence

KV2	Auth scoring 7 chiều theo graph context	Click essay → 94% “justified_growth” + 7-dim breakdown
KV3	Approach C linking: AI → Student → Mentor	Student add note → Mentor confirm → link becomes verified
KV4	Mentor dashboard với pending queue	Mentor thấy 3 items chờ verify, click confirm từng cái
KV5	QR badge scannable	Scan bằng điện thoại → ETESTERPublic load đúng
KV6	University requirements coverage	Grid 9 requirements, 8/9 covered ✓

3. Layer 1 — User View

Flow tổng thể

Student nộp artifact + optional context note
 ↓
 AI suggest trace links (với evidence)
 ↓
 Mentor nhận notification → review queue
 ↓
 Mentor: Confirm ✓ | Reject ✕ | Modify ~ (đổi relationship type)
 ↓
 Link confirmed → auth scoring chạy → core update → narrative rebuild
 ↓
 Manager: spot-check → issue badge → QR
 ↓
 University scan QR → verify page

Outputs

Artifact Graph — network diagram. Confirmed links = solid. Pending links = dashed. Click node → side panel.

Timeline — chronological. Click milestone → trace links + auth score + form + leadership.

AI Story — narrative EN + VN, rebuild after each confirmed link.

QR Badge — scan → public verify. No account needed.

Relationship types

Type	Meaning	Color
experience_source	Camp/CSR/award cung cấp experience cho essay	Teal
revision_of	Draft là bản sửa của draft trước	Amber

mentor_guided	Mentor session dẫn đến thay đổi	Purple
skill_applied	Kỹ năng học ở đây, dùng ở kia	Blue
score_progression	Test scores cải thiện theo thời gian	Gray
recommends	Mentor/teacher recommend học viên	Green

4. Linking Flow — Approach C

[AI] Đây là core flow của ETESTER. Implement đúng thứ tự. Không skip bước.

Tại sao Approach C

3 options đã được xét: - **Approach A** (Student tự link): Trust thấp — university không tin student tự certify. - **Approach B** (Mentor link thẳng): Mentor bottleneck, student mất agency. - **Approach C** (AI suggest → Student note → Mentor confirm): Trust cao nhất — AI + student memory + mentor expert verify cùng lúc.

AI không biết: intention của student khi link 2 artifacts

Student biết: context thật — "Tôi dùng PEEL từ camp này"

Mentor biết: cả 2 artifacts + session history → authority verify

Step-by-step flow

Step 1 — Student submit artifact

Student nộp bài/form xong → backend trigger Trace Engine ngay (async).

Step 2 — AI generates suggestions

Claude Trace Engine trả về top 3-5 suggested links với evidence. Saved vào MilestoneTraceLink với confirmed_by_mentor = False.

Step 3 — Student sees suggestions (ArtifactForm post-submit screen)

Màn hình hiện sau khi submit:

"AI phát hiện 3 kết nối có thể:"

- [] Writing Camp — Jul 2025 [confidence 94%]
 "Essay references thesis workshop from camp"
 Note (optional): _____
 placeholder: "VD: Tôi dùng cách viết thesis từ camp này..."
- [] Mentor Session #5 — Oct 2025 [confidence 91%]
 "Session focused on structure matches essay changes"
 Note (optional): _____
- [] CSR Share & Care — Aug 2025 [confidence 88%]

"Leadership narrative matches CSR coordinator role"
Note (optional): _____

[Bỏ qua tất cả] [Gửi cho mentor review →]

Rules: - Student KHÔNG confirm/reject — chỉ add note hoặc bỏ qua -
Tất cả suggestions đều được gửi cho mentor kèm student note (nếu có) - Student note được lưu vào
MilestoneTraceLink.student_context_note - Action này không block —
student có thể bỏ qua toàn bộ

Step 4 — Mentor receives notification

Mentor dashboard hiện badge: “3 items pending review”

Step 5 — Mentor reviews (MentorDashboard)

Mỗi pending link hiện:

Writing Camp → Personal Statement Draft #1

AI evidence:	"Essay references thesis workshop from camp"
Confidence:	94%
Student note:	"Tôi học cách viết thesis ở camp này và áp dụng vào bài"

[✓ Confirm] [✗ Reject] [~ Modify type ▼]

Mentor actions: - Confirm → confirmed_by_mentor = True, link becomes solid in graph - Reject → is_active = False, lưu MentorVerification với action trace_reject - Modify → đổi relationship_type rồi confirm

Step 6 — Post-confirm triggers

Sau mỗi mentor confirm: 1. auth_scoring.py chạy lại cho essay liên quan 2. ETESTERCore rebuild skills + stats 3. narrative_cache invalidated → rebuild async 4. Graph UI update: dashed edge → solid edge

5. Layer 2 — Dev + Tech

Stack

[AI] — ĐỌC CODEBASE TRƯỚC KHI LÀM BẤT KỲ THỨ GÌ

EESTER được build on top của Smart Parenting — cùng repo, cùng codebase. Trước khi viết bất kỳ dòng code nào, agent PHẢI:

1. Đọc requirements.txt hoặc pyproject.toml → detect backend stack + AI SDK đang dùng
2. Đọc package.json → detect frontend stack + libraries đang dùng
3. Đọc shared/model.py (đã có trong context) → biết models nào đã có
4. Đọc cấu trúc thư mục hiện tại → hiểu pattern tổ chức code
5. **Dùng đúng những gì đã có.** Không install thư viện mới nếu

đã có thứ tương đương.

Ví dụ: nếu Smart Parenting dùng openai SDK → ETESTER dùng openai luôn, không dùng anthropic. Nếu dùng pyjwt → dùng pyjwt. Nếu dùng axios → dùng axios. Không assume. Đọc rồi mới code.

Chỉ những thứ sau là ETESTER thêm mới — chưa có trong Smart Parenting:

Graph UI: SVG thuần – artifact network diagram, không cần thư viện
QR code: react-qr-code – generate QR badge
JWT sign: dùng JWT library đang có sẵn trong codebase (detect từ requirements)

Production only — không build trong hackathon:

W3C VC: didkit – Ed25519 credential signing

DB priority

PRIORITY 1 – Reuse từ model.py (KHÔNG SỬA):
Student, Mentor, Parent, Milestone, BehavioralLog,
Conversation, School, Course, User, TimestampMixin

PRIORITY 2 – Extend qua bảng liên kết:
MilestoneTraceLink (thêm student_context_note field)
MilestoneArtifact

PRIORITY 3 – Thực thể mới:
ArtifactForm · MentorVerification · AuthScoringResult
ETESTERCore · ETESTERBadge

API Routes

| **[AI]** Implement đúng list này. Không thêm route ngoài list.

```
# Student endpoints
GET /api/etester/:studentId
  → ETESTERCore + milestones + trace_links + auth_results +
  narrative

POST /api/etester/contribute
  body: { studentId, milestoneId, formData, artifactText }
  → save ArtifactForm + MilestoneArtifact
  → async: trigger Trace Engine → save suggested links
  → return: { milestone, suggested_links[] }

POST /api/etester/student-note/:traceLinkId
  body: { studentNote: string }
  → update MilestoneTraceLink.student_context_note
  → notify mentor (optional notification)

# Mentor endpoints
GET /api/etester/mentor/:mentorId/pending
  → list pending trace links + artifacts for this mentor's
  students
  → return: { pending_links[], pending_approvals[] }

POST /api/etester/verify-trace/:traceLinkId
```



```

    body: { mentorId, action: "confirm"|"reject"|"modify",
              new_relationship_type?: string, note?: string }
    → update MilestoneTraceLink
    → save MentorVerification
    → if confirm: trigger auth-score + rebuild-core

POST /api/etester/approve-artifact/:milestoneId
    body: { mentorId, approved: bool, note?: string }
    → update Milestone.mentor_approved
    → save MentorVerification

# AI endpoints (backend triggers, not direct frontend calls)
POST /api/etester/trace/:milestoneId          (internal async)
POST /api/etester/score-auth/:milestoneId      (internal, after
verify-trace)
POST /api/etester/rebuild-narrative/:studentId (internal, after
score-auth)

# Manager endpoints
POST /api/etester/issue-badge/:studentId
    body: { managerId }
    → sign JWT → save ETESTERBadge → return badge_uid + QR URL

# Public
GET /api/etester/verify/:badgeUid              (no auth)

```

File structure

```

backend/
├── shared/
│   ├── model.py                # Smart Parenting (KHÔNG SỬA)
│   └── etester_models.py       # ETESTER models
├── api/etester/
│   ├── routes.py
│   ├── services.py
│   ├── claudeservice.py        # 4 Claude prompts
│   └── auth_scoring.py        # 7-dimension engine
└── alembic/versions/002_etester.py

frontend/src/
├── pages/
│   ├── ETESTERDashboard.jsx    # Student view
│   ├── MentorDashboard.jsx     # Mentor view – NEW
│   └── ETESTERPublic.jsx       # QR destination
├── components/etester/
│   ├── ArtifactGraph.jsx
│   ├── MilestoneTimeline.jsx
│   └── ArtifactForm.jsx        # Includes post-submit
├── suggestion screen
│   └── TraceLinkSuggestions.jsx # Post-submit: student sees AI
├── suggestions – NEW
│   ├── MentorPendingQueue.jsx  # Mentor review queue – NEW
│   ├── MentorTraceReview.jsx   # Single link review card – NEW
│   ├── NarrativeCard.jsx
│   ├── QRBadge.jsx
│   ├── AuthScoreCard.jsx
│   └── ContributorDonut.jsx
└── services/etesterApi.js

```

6. Layer 3 — AI Vibe Code

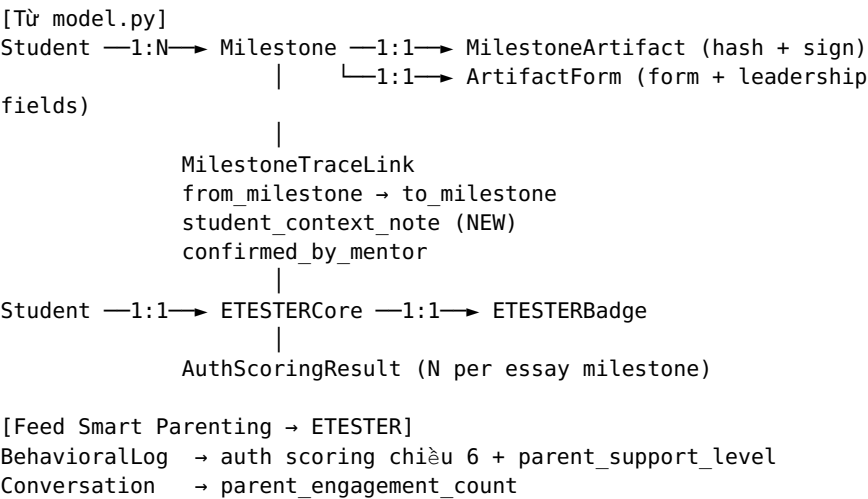
| [AI] Build theo thứ tự. Mỗi step phụ thuộc step trước.

Step 1	etester_models.py	7 models + student_context_note field
Step 2	alembic migration	python -m alembic revision --autogenerate
Step 3	Seed data	18–20 milestones + links (confirmed=True) + forms
Step 4	Claude prompts	Test 4 prompts, verify JSON format
Step 5	API: GET /etester/:id	Full data response
Step 6	API: POST /contribute	+ async trace trigger
Step 7	API: Mentor pending + verify-trace	
Step 8	ArtifactGraph.jsx	Solid vs dashed edges
Step 9	TraceLinkSuggestions.jsx	Post-submit screen với student notes
Step 10	MentorDashboard.jsx	Pending queue + review cards
Step 11	MilestoneTimeline.jsx	+ trace panel + auth breakdown
Step 12	AuthScoreCard.jsx	7-dim breakdown
Step 13	NarrativeCard.jsx	+ regenerate
Step 14	QRBadge + ETESTERPublic	Scan test on real phone

7. Database Models

| [AI] File: shared/etester_models.py

Quan hệ tổng thể



MilestoneTraceLink

```
class MilestoneTraceLink(Base, TimestampMixin):
    __tablename__ = "milestone_trace_links"

    id: int (PK)
    from_milestone_id: int FK → milestones.id (RESTRICT)
    to_milestone_id: int FK → milestones.id (RESTRICT)
```

```

student_id:          str FK → students.student_id (CASCADE)

relationship_type:    str
# experience_source | revision_of | mentor_guided
# skill_applied | score_progression | recommends

text
evidence:            Text | None    # AI-generated evidence
confidence:           float          # 0.0–1.0 (AI
confidence)

# Approach C fields
explanation
student_context_note: Text | None    # Student's optional
student_noted_at:      datetime | None

# Mentor verification
suggested_by_ai:       bool = True
confirmed_by_mentor:    bool = False
confirmed_by_mentor_id: int FK → mentors.mentor_id | None
confirmed_at:           datetime | None

is_active:             bool = True
# Rule: confirmed=True → is_active cannot be set False
# Enforce at application layer

# Constraints:
# UniqueConstraint(from_milestone_id, to_milestone_id)
# CheckConstraint confidence 0.0–1.0
# CheckConstraint relationship_type IN (6 types)

```

MilestoneArtifact

```

class MilestoneArtifact(Base):
    __tablename__ = "milestone_artifacts"

    id:                int (PK)
    milestone_id:       int FK → milestones.id UNIQUE
    student_id:         str FK → students.student_id

    full_text_content:  Text | None    # AI reads for trace + auth
    artifact_hash:      str(64) | None # SHA256
    prev_artifact_hash: str(64) | None

    # 3-stage signing
    vc_signature:       Text | None
    mentor_signed:      bool = False
    admin_signed:       bool = False
    manager_signed:     bool = False
    signed_at:          datetime | None

    created_at:         datetime

```

ArtifactForm

```

class ArtifactForm(Base, TimestampMixin):
    __tablename__ = "artifact_forms"

```

```

id: int (PK)
milestone_id: int FK → milestones.id UNIQUE
student_id: str FK → students.student_id

activity_type: str # See Section 8
form_version: str = "1.0"
form_data: JSONB = {} # Schema per type, see

```

Section 8

```

# Derived – denormalized for fast query
blooms_level: str | None
skills_practiced: JSONB = []

# Leadership – dedicated columns for direct query + aggregate
had_leadership_role: bool = False
leadership_role_title: str | None
leadership_team_size: int | None
leadership_outcome: str | None # max 100 words,
measurable

# Mentor comment – mentor fills, student cannot edit
mentor_comment: Text | None
mentor_comment_by: int FK → mentors.mentor_id | None

```

MentorVerification (append-only)

```

class MentorVerification(Base):
    __tablename__ = "mentor_verifications"
    # NO updated_at. NO DELETE.

    id: int (PK)
    milestone_id: int FK → milestones.id (RESTRICT)
    student_id: str FK → students.student_id (CASCADE)

    verifier_id: int
    verifier_type: str # "mentor" | "admin" | "manager"

    action_type: str
    # mentor_approve | mentor_reject | admin_verify | admin_flag
    # trace_confirm | trace_reject | trace_modify

    note: Text | None
    trace_link_id: int FK → milestone_trace_links.id | None

    created_at: datetime (immutable)

```

AuthScoringResult

```

class AuthScoringResult(Base):
    __tablename__ = "auth_scoring_results"

    id: int (PK)
    milestone_id: int FK → milestones.id (CASCADE)
    student_id: str FK → students.student_id (CASCADE)

    auth_score: int # 0–100

```

```

verdict:                str
# justified_growth | consistent | needs_review
# suspicious_jump  | insufficient_data

dimension_scores:        JSONB = {}
# { "d1": { score, weight, note }, ... "d7": { score, weight,
note } }

# JSONB → add new dimensions without migration

explaining_artifacts:    JSONB = []
# [{ id, relationship, explains }]

explanation_en:           Text | None
explanation_vn:           Text | None
graph_snapshot:          JSONB = {}

scored_at:               datetime

```

EESTERCore

```

class EESTERCore(Base, TimestampMixin):
    __tablename__ = "etestester_core"

    id:                    int (PK)
    student_id:            str FK UNIQUE

    academic_score:        Numeric(5,2) | None
    writing_growth:         Numeric(5,2) | None

    skills:                JSONB = []
    # [{
    #   skill, evidence_count, artifacts[],
    #   had_formal_role (leadership),
    #   max_team_size, ai_summary
    # }]

    mentor_verifications:  int = 0
    total_contributions:   int = 0
    contributor_breakdown: JSONB = {}

    # Pending counts – for mentor badge notification
    pending_trace_links:   int = 0
    pending_approvals:     int = 0

    requirements_coverage: JSONB = {}
    # { english_proficiency, standardized_test, personal_essay,
    #   extracurricular, community_service, leadership_evidence,
    #   academic_record, recommendation_context, awards_honors }

    parent_support_level:  str = "low"
    parent_engagement_count: int = 0
    institutional_stamp:    bool = False
    stamped_at:            datetime | None
    stamped_by:            int FK → mentors.mentor_id | None
    consistency_score:     int = 0

    narrative_en:          Text | None
    narrative_vn:          Text | None
    narrative_updated_at:   datetime | None

```

badge_issued: bool = False

ETESTERBadge

```
class ETESTERBadge(Base):
    __tablename__ = "etester_badges"

    id: int (PK)
    core_id: int FK → etester_core.id UNIQUE
    student_id: str FK → students.student_id

    badge_uid: str UNIQUE # "ETR-2026-001234"
    credential_type: str = "jwt_rs256"
    signed_token: Text
    issuer_did: str = "did:web:etest.edu.vn"

    badge_payload: JSONB = {}
    # name, program, months, milestones, mentor_verified,
    # ielts, sat, auth_avg, essays_approved,
    # leadership_evidence, requirements_coverage

    issued_at: datetime
    expires_at: datetime | None
    is_revoked: bool = False
    revoked_at: datetime | None
    revoke_reason: Text | None
    verify_count: int = 0
    last_verified_at: datetime | None
```

8. Artifact Types + Form Schemas

[AI] activity_type values và form_data JSONB schema tương ứng.
Leadership fields (had_leadership_role, leadership_role_title, leadership_team_size, leadership_outcome) là **dedicated columns** trong ArtifactForm — KHÔNG nằm trong form_data.

ielts_mock / sat_mock / toefl_mock

```
{
  "score_this": 6.5,
  "score_prev": 6.0,
  "score_breakdown": { "listening": 7.0, "reading": 7.0,
                       "writing": 6.0, "speaking": 6.5 },
  "test_center": "ETEST Q3",
  "official": false,
  "weak_areas": ["writing task 1"],
  "strategy_note": "Focus on data description vocabulary"
}
```

Leadership: N/A

essay_draft / essay_final

```
{
```

```
    "draft_number": 2,
    "topic": "Personal Statement",
    "word_count": 650,
    "target_school": "University of Melbourne",
    "improvement_from_prev": "Stronger thesis, added counter-
argument",
    "forward_intent": "Will use in Common App"
  }
```

Leadership: N/A

camp

```
{
  "camp_name": "ETEST Writing Camp",
  "organizer": "ETEST Vietnam",
  "duration_days": 21,
  "location": "HCMC",
  "key_skills": ["thesis writing", "PEEL structure"],
  "key_experience": "Led morning discussion sessions",
  "forward_intent": "Apply PEEL in Personal Statement"
}
```

Leadership columns: had_leadership_role=true,
leadership_role_title="Morning session facilitator",
leadership_team_size=8, leadership_outcome="Group improved draft
quality by end of week 2"

csr

```
{
  "org_name": "Share and Care",
  "activity_type": "tutoring",
  "duration_weeks": 12,
  "hours_per_week": 3,
  "beneficiaries": "underprivileged students",
  "beneficiary_count": 15,
  "impact_description": "Helped 15 students pass grade 5 math exam"
}
```

Leadership columns: had_leadership_role=true,
leadership_role_title="Volunteer coordinator",
leadership_team_size=5, leadership_outcome="Organized 3 community
events, 50+ attendees"

extracurricular

```
{
  "org_name": "English Club THPT Gia Dinh",
  "role": "Vice President",
  "duration_months": 8,
  "commitment_hrs_per_week": 4,
  "description": "Organized weekly debates and English movies
night",
  "achievements": "Grew membership from 20 to 45 in one semester"
}
```

Leadership columns: had_leadership_role=true,
leadership_role_title="Vice President", leadership_team_size=45,
leadership_outcome="Grew club 2x, organized 12 events"

mentor_session

```
{  
  "focus_area": "Personal Statement structure",  
  "action_items": ["Rewrite opening hook", "Add specific example"],  
  "readiness_level": 4,  
  "student_note": "Finally understood how to open with a scene"  
}
```

Leadership: N/A

consultation

```
{  
  "topic": "School list finalization",  
  "schools_discussed": ["Melbourne", "RMIT", "Monash"],  
  "decisions_made": "Set Melbourne as EA target",  
  "next_steps": ["Complete supplements by Nov 1"]  
}
```

Leadership: N/A

recommendation_letter

```
{  
  "recommender_name": "Nguyen Van Cuong",  
  "recommender_role": "AMP Mentor",  
  "recommender_org": "ETEST Vietnam",  
  "relationship": "14-month AMP mentor",  
  "relationship_duration_months": 14,  
  "submitted_to": ["University of Melbourne", "RMIT"],  
  "date_sent": "2026-01-15",  
  "key_themes": ["intellectual curiosity", "leadership",  
"resilience"],  
  "letter_excerpt": "Optional – mentor cho phép share 1-2 sentences"  
}
```

Leadership: N/A

academic_record

```
{  
  "school_name": "THPT Gia Dinh",  
  "semester": "2024-2025 HK1",  
  "grade_system": "10-point",  
  "subjects": [  
    { "name": "Toán", "grade": 9.2, "credit_hours": 4 },  
    { "name": "Văn", "grade": 8.8, "credit_hours": 3 },  
    { "name": "Anh", "grade": 9.5, "credit_hours": 3 }  
  ],  
  "gpa_this_term": 9.1,  
}
```



```
      "cumulative_gpa": 9.0,  
      "rank_in_class": "3/45"  
    }  
  ]  
}
```

Note: Data imbalance across grade systems is intentional and acceptable. Store raw. Do not normalize. University reads as-is.

Leadership: N/A

award_honor

```
{  
  "award_name": "First Prize – National English Essay Competition",  
  "issuing_org": "Ministry of Education Vietnam",  
  "level": "national",  
  "year": 2025,  
  "category": "academic",  
  "description": "Essay on climate change and youth responsibility",  
  "certificate_issued": true  
}
```

Leadership: N/A

9. Artifact Graph

[AI] Component: ArtifactGraph.jsx

Node visual rules

Node type	Shape	Color
essay_draft, essay_final	Rounded rect	Amber
camp, csr, extracurricular, award_honor	Circle	Teal
ielts_mock, sat_mock	Circle	Gray
mentor_session, recommendation_letter	Rounded rect	Purple
academic_record	Rounded rect	Blue
ETESTERCore	Large rounded rect filled	Purple dark

Node badges: - mentor_approved = true → thick border 2px -
Unconfirmed links TO this node → small amber dot top-right “⚠” -
had_leadership_role = true → small “L” badge bottom-right

Edge visual rules

State	Style
confirmed_by_mentor = true	Solid stroke, full opacity
confirmed_by_mentor = false	Dashed stroke, opacity 0.4, amber dot at midpoint
revision_of	Thicker 2.5px
contributes_to (→ Core)	Blue dashed

Interaction

Default: all nodes + edges visible
Click node: connected nodes → opacity 1.0
unconnected → opacity 0.15
connected edges → solid + show label
unconnected edges → opacity 0.08
Click again: reset

Side panel on click:
Title + date + type badge
Auth score chip (color by verdict) – essays only
Leadership badge if had_leadership_role
Trace links list:
Each: relationship chip + confidence% + status
confirmed: "✓ Mentor verified" green
pending: "⌘ Awaiting mentor" amber
student_note (if exists): italic below link
"View in timeline →" button

Seed trace links (pre-confirmed for demo)

```
SEED_TRACE_LINKS = [  
  { "from": "camp_001", "to": "essay_draft_001",  
    "type": "experience_source", "confidence": 0.94,  
    "evidence": "Essay draft references Writing Camp thesis  
workshop",  
    "student_context_note": "Tôi học cách viết thesis ở camp và áp  
dụng ngay vào Draft #1",  
    "confirmed_by_mentor": True },  
  
  { "from": "csr_003", "to": "essay_draft_001",  
    "type": "experience_source", "confidence": 0.88,  
    "evidence": "Leadership narrative matches CSR coordinator role",  
    "student_context_note": "Câu chuyện leadership trong essay lấy  
từ trải nghiệm CSR",  
    "confirmed_by_mentor": True },  
  
  { "from": "session_005", "to": "essay_draft_001",  
    "type": "mentor_guided", "confidence": 0.91,  
    "evidence": "Session focused on structure – draft shows new PEEL  
format",  
    "confirmed_by_mentor": True },  
  
  { "from": "essay_draft_001", "to": "essay_draft_002",  
    "type": "revision_of", "confidence": 0.97,  
    "evidence": "Opening preserved, body restructured per session  
notes",  
    "confirmed_by_mentor": True },  
  
  { "from": "session_007", "to": "essay_draft_002",  
    "type": "mentor_guided", "confidence": 0.93,  
    "evidence": "Counter-argument added per session feedback",  
    "confirmed_by_mentor": True },  
  
  { "from": "essay_draft_002", "to": "essay_final_001",  
    "type": "revision_of", "confidence": 0.98,  
    "evidence": "Final preserves Draft 2 structure, refined intro",  
    "confirmed_by_mentor": True },
```

```

{ "from": "session_007", "to": "essay_final_001",
  "type": "mentor_guided", "confidence": 0.89,
  "evidence": "Mentor approved after final session review",
  "confirmed_by_mentor": True },

{ "from": "rec_letter_001", "to": "essay_final_001",
  "type": "recommends", "confidence": 1.0,
  "evidence": "Mentor formally recommends to target universities",
  "confirmed_by_mentor": True },

{ "from": "award_001", "to": "essay_final_001",
  "type": "experience_source", "confidence": 0.82,
  "evidence": "Essay references national competition experience",
  "student_context_note": "Tôi viết về bài thi quốc gia trong
essay",
  "confirmed_by_mentor": True },
]

```

10. Auth Scoring Framework

[AI] File: backend/api/etester/auth_scoring.py
 Prerequisite: Pull confirmed trace_links TRƯỚC khi chạy. Auth needs graph data.
 Trigger: After each verify-trace confirm action.

Core principle

“Bài này có fit vào toàn bộ hành trình của học viên này không?”

Không hỏi “AI viết bài này không?” — hỏi “Growth này có được giải thích bởi graph không?”

7 Dimensions

#	Dimension	Weight	Key question
D1	Growth explained by artifacts	0.25	Có artifact linked giải thích growth không?
D2	Topic familiarity matches history	0.15	Vocabulary/topic match artifact history?
D3	Narrative timeline consistent	0.15	Events referenced có artifacts prior không?
D4	Error pattern consistent	0.15	Errors giảm dần hay biến mất đột ngột?
D5	Mentor notes corroborate	0.20	Mentor notes align với essay quality?
	Behavioral		Study hours/streak

D6	context matches	0.05	match submission quality?
D7	Structural fingerprint consistent	0.05	Writing structure consistent với history?

D1 — Growth explained (most important)

Situation	Verdict
Score tăng + artifact linked (confidence ≥ 0.7 , mentor confirmed)	justified_growth
Score tăng + mentor session note aligned	justified_growth
Score ổn định qua nhiều bài	consistent
Score nhảy + không artifact nào explain	suspicious_jump
Chưa đủ writing samples (< 2)	insufficient_data

Rule tuyệt đối: Nếu explaining_artifacts có ít nhất 1 artifact với confidence ≥ 0.7 VÀ confirmed_by_mentor = True VÀ D5 $\geq 80 \rightarrow$ verdict PHẢI là justified_growth. Không được flag suspicious_jump.

Verdict logic

```
def compute_verdict(
    dimension_scores: dict,
    explaining_artifacts: list,
    writing_sample_count: int
) -> str:
    if writing_sample_count < 2:
        return "insufficient_data"

    weighted = sum(d["score"] * d["weight"]
                   for d in dimension_scores.values())

    d1 = dimension_scores["d1_growth_explained"]
    d5 = dimension_scores["d5_mentor_corroborate"]["score"]

    # Justified growth: large increase but explained
    if d1.get("growth_delta", 0) > 10 and explaining_artifacts:
        if d5 >= 80:
            return "justified_growth"

    if weighted >= 85: return "consistent"
    if weighted >= 65: return "needs_review"
    if d1.get("suspicious") and not explaining_artifacts:
        return "suspicious_jump"
    return "needs_review"
```

Output format

```
{
```

```

    "auth_score": 94,
    "verdict": "justified_growth",
    "dimension_scores": {
      "d1_growth_explained": { "score": 95, "weight": 0.25,
        "growth_delta": 16,
        "note": "Growth +16pts explained by Writing Camp (confidence
0.94, mentor confirmed)" },
      "d2_topic_familiarity": { "score": 92, "weight": 0.15,
        "note": "Leadership claims match camp role + CSR coordinator
artifact" },
      "d3_narrative_timeline": { "score": 90, "weight": 0.15,
        "note": "All referenced events have prior artifacts with
correct dates" },
      "d4_error_pattern": { "score": 88, "weight": 0.15,
        "note": "Comma splice reduced gradually – consistent with
mentor feedback" },
      "d5_mentor_corroborate": { "score": 96, "weight": 0.20,
        "note": "Mentor session #7 improvement aligns with final essay
quality" },
      "d6_behavioral_context": { "score": 85, "weight": 0.05,
        "note": "3.2h/day average 2 weeks before submission" },
      "d7_structural_fingerprint": { "score": 91, "weight": 0.05,
        "note": "Paragraph opening style consistent with all previous
drafts" }
    },
    "explaining_artifacts": [
      { "id": "camp_001", "relationship": "experience_source",
        "explains": "vocabulary_expansion + leadership narrative",
        "student_note": "Tôi học cách viết thesis ở camp và áp dụng
vào bài" }
    ],
    "explanation_en": "Score increase from 78% to 94% is fully
explained by verified artifacts: Writing Camp (Jul 2025) provided
vocabulary foundation; Mentor Session #7 guided counter-argument
development. Growth trajectory is consistent with expected learning
progression.",
    "explanation_vn": "Điểm tăng từ 78% lên 94% được giải thích đầy đủ
bởi các artifact đã xác nhận: Trại hè Writing cung cấp nền tảng từ
vựng; Buổi học mentor #7 hướng dẫn lập luận phản biện."
  }
}

```

Extensibility note

dimension_scores là JSONB — thêm dimension mới chỉ cần update Claude prompt + weight distribution. Không cần DB migration.

11. Claude Prompts

[AI] File: backend/api/etester/claude_service.py

Helper

```

import anthropic, json
from typing import Any

client = anthropic.Anthropic()

```

```

async def call_claude(prompt: str, max_tokens: int = 800) -> str:
    try:
        msg = client.messages.create(
            model="claude-sonnet-4-6",
            max_tokens=max_tokens,
            messages=[{"role": "user", "content": prompt}]
        )
        return msg.content[0].text
    except Exception as e:
        print(f"Claude error: {e}")
        return "{}"

def parse_json(text: str, fallback: Any = None) -> Any:
    try:
        clean = text.replace("`json", "").replace("`", "").strip()
        return json.loads(clean)
    except Exception:
        return fallback if fallback is not None else {}

```

Prompt 1 — Trace Engine

```

async def trace_artifact(
    new_milestone: dict,
    artifact_history: list[dict],
    form_data: dict | None = None
) -> list[dict]:

    top5 = sorted(
        [a for a in artifact_history if a["date"] <
new_milestone["date"]],
        key=lambda x: x["date"]
    )[-5:]

    form_ctx = f"\nForm data: {json.dumps(form_data)}" if form_data
else ""

    prompt = f"""
You are analyzing learning artifact connections for a Vietnamese
student
building a verified study portfolio (ETESTER).

NEW ARTIFACT:
- type: {new_milestone['type']}
- title: {new_milestone['title']}
- date: {new_milestone['date']}
- summary: {new_milestone.get('ai_summary', 'N/A')}
{form_ctx}

RECENT HISTORY (up to 5 prior artifacts):
{chr(10).join([
    f"- id:{a['id']} [{a['date']}] ({a['type']}) {a['title']}:
{a.get('ai_summary', '')}"
    for a in top5
])}

Find CAUSAL relationships only – not just "happened before" but
"actually contributed to".
Only return links with genuine evidence from content or form data.

```

```

Relationship types: experience_source | revision_of | mentor_guided
|
| skill_applied | score_progression | recommends

Return JSON only – empty array if no genuine links:
[{{
  "from_milestone_id": "id",
  "relationship_type": "type",
  "evidence": "specific observation",
  "confidence": 0.0-1.0
}}]
"""

result = await call_claude(prompt, max_tokens=600)
return parse_json(result, fallback=[])

```

Prompt 2 — Auth Scorer (7 dimensions)

```

async def score_auth_7dim(
    new_essay_text: str,
    writing_samples: list[dict],
    confirmed_trace_links: list[dict],
    mentor_notes: list[str],
    behavioral_stats: dict,
    student_name: str,
    leadership_artifacts: list[dict]
) -> dict:

    explaining = [
        t for t in confirmed_trace_links
        if t["relationship_type"] in ["experience_source",
"mentor_guided"]
        and t["confidence"] >= 0.7
        and t["confirmed_by_mentor"]
    ]

    # Include student context notes in explaining artifacts
    explaining_with_notes = [
        { **t, "student_note": t.get("student_context_note", "") }
        for t in explaining
    ]

    prompt = f"""
You are scoring writing authenticity for {student_name}.

CORE PRINCIPLE: Ask "Does this essay fit this student's complete
journey?"
NOT "Did AI write this?"

IMPORTANT RULES:
1. If explaining_artifacts exist (confidence ≥ 0.7, mentor
confirmed) AND
D5 (mentor corroborate) ≥ 80 → verdict MUST be "justified growth"
2. Student context notes (their own explanation) are strong evidence
3. Vietnamese L1 interference patterns are authentic markers – do
not penalize
4. Gradual error reduction with mentor sessions = expected growth

HISTORICAL WRITING:

```

```

        {chr(10).join([f"[{s['date']}] ({s['type']}): {s['content'][:400]}"
for s in writing_samples])}

NEW ESSAY:
{new_essay_text[:1200]}

CONFIRMED TRACE LINKS (mentor-verified artifacts that explain this
essay):
    {chr(10).join([
        f"- {t.get('from_title','')} ({t['relationship_type']}, conf
{t['confidence']}): {t['evidence']}"
        + (f"\n Student note: {t['student_note']}" if
t.get('student_note') else "")
        for t in explaining_with_notes
    ]) or "None confirmed yet"}

MENTOR NOTES (from linked sessions):
{chr(10).join(mentor_notes) or "No linked mentor notes"}

BEHAVIORAL CONTEXT:
avg_hours/day: {behavioral_stats.get('avg_hours','')}
streak: {behavioral_stats.get('streak','')} days
late_nights: {behavioral_stats.get('late_nights', 0)}

LEADERSHIP CONTEXT:
    {chr(10).join([f"- {a['title']}: {a.get('leadership_role_title')},
{a.get('leadership_team_size')} ppl" for a in leadership_artifacts])
or "None documented"}

Score each dimension 0-100:
d1_growth_explained (0.25) – growth explained by graph artifacts?
d2_topic_familiarity (0.15) – topic knowledge matches artifact
history?
d3_narrative_timeline (0.15) – essay references have prior
artifacts?
d4_error_pattern (0.15) – error patterns consistent with history?
d5_mentor_corroborate (0.20) – mentor notes align with essay
quality?
d6_behavioral_context (0.05) – study behavior supports submission?
d7_structural_fingerprint (0.05) – writing structure consistent?

Return JSON only:
{{
    "auth_score": 0-100,
    "verdict":
"justified_growth|consistent|needs_review|suspicious_jump|insufficient_data",

    "dimension_scores": {{
        "d1_growth_explained":        {"score":0-
100,"growth_delta":0,"note":"..."}},
        "d2_topic_familiarity":        {"score":0-100,"note":"..."}},
        "d3_narrative_timeline":        {"score":0-100,"note":"..."}},
        "d4_error_pattern":            {"score":0-100,"note":"..."}},
        "d5_mentor_corroborate":        {"score":0-100,"note":"..."}},
        "d6_behavioral_context":        {"score":0-100,"note":"..."}},
        "d7_structural_fingerprint":    {"score":0-100,"note":"..."}},
    }},
    "explanation_en": "1-2 sentences",
    "explanation_vn": "1-2 câu tiếng Việt"
}}
```



```

"""
result = await call_claude(prompt, max_tokens=1000)
return parse_json(result, fallback={
    "auth_score": 85, "verdict": "insufficient_data",
    "dimension_scores": {},
    "explanation_en": "Insufficient data for full analysis.",
    "explanation_vn": "Chưa đủ dữ liệu."
})

```

Prompt 3 — Narrative Builder

```

async def build_narrative(
    student: dict,
    milestones: list[dict],
    etester_core: dict
) -> dict:

    completed = [m for m in milestones if m["status"] ==
"completed"]
    leadership = [m for m in completed if m.get("form",
{}).get("had_leadership_role")]

    prompt = f"""
        Build a verified learning credential narrative for
        {student['name']},
        {student['months_enrolled']} months in {student['program']} at ETEST
        Vietnam.

        Verified milestones (chronological):
        {chr(10).join([
            f"- {m['date']}: [{m['type']}] {m['title']}"
            + (f" score:{m['score']}" if m.get('score') else "")
            + (f" auth:{m['auth_score']}" if m.get('auth_score') else "")
            + (" ✓mentor" if m.get('mentor_approved') else "")
            for m in completed
        ])}

        Leadership evidence:
        {chr(10).join([f"- {m['title']}: {m['form']}"
        ['leadership_role_title']] (m['form']['leadership_team_size']
        people) → {m['form']['leadership_outcome']} for m in leadership))
        or "None documented"}

        Requirements covered:
        {json.dumps(etester_core.get('requirements_coverage', {}))}

        Write TWO paragraphs:
        1. English (university): Professional, evidence-based. Reference
        actual dates.
        Show progression. Highlight leadership if present. No generic
        phrases.
        2. Vietnamese (parents): Warm, plain language. 30-second read. No
        jargon.

        Return JSON: {"narrative_en": "...", "narrative_vn": "..."}
        """

    result = await call_claude(prompt, max_tokens=900)
    return parse_json(result, fallback={
        "narrative_en": "Narrative being generated...",

```

```
        "narrative_vn": "Đang tạo..."
    })
```

Prompt 4 — Leadership Aggregator

```
async def aggregate_leadership(
    student_name: str,
    leadership_artifacts: list[dict]
) -> dict:

    if not leadership_artifacts:
        return {"has_leadership": False}

    prompt = f"""
    Synthesize leadership evidence for {student_name} from verified
    activities.

    Activities (all mentor/admin confirmed):
    {chr(10).join([
        f"- [{a['date']}] {a['title']}: role=
    {a.get('leadership_role_title')}, "
        f"team={a.get('leadership_team_size')}, outcome=
    {a.get('leadership_outcome')}"
        for a in leadership_artifacts
    ])}

    Be specific and evidence-based. Do not overstate.

    Return JSON:
    {{
        "has_leadership": true,
        "evidence_count": {len(leadership_artifacts)},
        "max_team_size": <max>,
        "duration_months": <span first to last>,
        "ai_summary": "2 sentences, specific",
        "key_roles": ["role 1", "role 2"],
        "credibility": "high|medium|low"
    }}
    """

    result = await call_claude(prompt, max_tokens=400)
    return parse_json(result, fallback={"has_leadership": False})
```

12. Component Specs

[AI] All in frontend/src/components/etester/ and
frontend/src/pages/

ArtifactGraph.jsx

Props:

- milestones: Milestone[]
- traceLinks: TraceLink[] # includes student_context_note,
- confirmed_by_mentor
- onNodeSelect: (id) => void

Node badges:

mentor_approved=true → thick border 2px
has pending unconfirmed links → amber dot "⌘" top-right
had_leadership_role=true → "L" badge bottom-right

Edge states:

confirmed_by_mentor=true → solid, full opacity
confirmed_by_mentor=false → dashed 0.4 opacity + amber midpoint dot

Side panel on click:

Title + date + type
Auth score chip (essays only)
Leadership summary (if had_leadership_role)
Trace links list:
relationship chip + confidence%
"✓ Mentor verified" green OR "⌘ Awaiting mentor" amber
student_context_note in italic below (if exists)
"View in timeline →"

TraceLinkSuggestions.jsx — NEW

Shown immediately after student submits artifact (post-submit screen).

Props:

suggestions: TraceLink[] # AI-suggested, unconfirmed
onNotesSubmit: (notes: {[traceLinkId]: string}) => void
onSkip: () => void

Layout:

Header: "AI phát hiện {N} kết nối có thể"
Subtext: "Xem lại và thêm ghi chú nếu muốn – mentor sẽ xác nhận"

For each suggestion:

[teal dot] Writing Camp – Jul 2025 relationship: experience_source confidence: 94%
AI: "Essay references thesis workshop"
Ghi chú của bạn (tùy chọn): [_____] VD: Tôi học cách viết thesis ở camp này...

Bottom actions:

[Bỏ qua tất cả] [Gửi cho mentor review →]

Rules:

- Student does NOT confirm/reject links – only adds optional note
- "Bỏ qua" sends all suggestions to mentor WITHOUT notes
- "Gửi" sends suggestions WITH notes to mentor
- Either action saves notes to

MilestoneTraceLink.student_context_note

- Non-blocking: student can leave page anytime
-

MentorDashboard.jsx — NEW

Main page for mentor. Route: /mentor/dashboard

Props: none (uses auth context for mentorId)

Layout:

Header: "Mentor Dashboard" + mentor name + "Powered by ETEST"

Pending queue section:

Title: "Cần xác nhận ({total_pending})"

Two tabs:

[Trace Links ({N})] [Artifact Approvals ({N})]

Tab 1 – Trace Links:

List of MentorTraceReview cards (see below)

Sorted by: student name → date submitted

Tab 2 – Artifact Approvals:

List of pending milestone approvals

Each: student name + milestone title + date + type badge

[✓ Approve] [✗ Reject] + optional note input

My students section:

Summary cards per student:

Student name + program + total contributions

Progress bar (EESTERCore.total_contributions)

"View ETESTER →" link

MentorTraceReview.jsx — NEW

Single trace link review card. Used inside MentorDashboard.

Props:

traceLink: MilestoneTraceLink (with from/to milestone details)

onConfirm: (id, newType?) => void

onReject: (id, note) => void

Layout:

```
[Student name] · submitted {relative date}

Writing Camp (Jul 2025)
  —[experience_source]→
Essay Draft #1 (Oct 2025)

AI evidence:
"Essay references thesis workshop from camp"
Confidence: 94%

Student note:
"Tôi học cách viết thesis ở camp và áp dụng..."

Relationship: [experience_source ▼] (editable)

[✓ Confirm]    [✗ Reject]
Note (optional): [_____]
```

Rules:

- Relationship type dropdown is editable before confirm (Modify flow)
 - Confirm saves: `confirmed_by_mentor=True`, `relationship_type` (possibly changed)
 - Reject saves: `is_active=False` + `MentorVerification` record
 - Both save `MentorVerification` with action `trace_confirm` or `trace_reject`
 - After confirm: trigger `POST /api/etester/score-auth/:milestoneId` (async)
-

MilestoneTimeline.jsx

Props:

```
milestones: Milestone[]
traceLinks: TraceLink[]
authResults: AuthScoringResult[]
```

Filter tabs: All | Essay | Academic | Activities | Verified

Each milestone row:

```
Colored dot (contributor_type)
Title + relative date
Score badge (if exists)
Auth score chip (essays: color by verdict)
"✓ Mentor" tag (if mentor_approved)
"L" leadership badge (if had_leadership_role)
"⌕ {N} pending" badge (if unconfirmed trace links)
```

Click → side panel:

```
Full details + form_data summary
Leadership fields (if applicable)
Trace links with status + student notes
Auth score breakdown (if essay)
```

AuthScoreCard.jsx

Props: `authResult: AuthScoringResult`

```
Large score circle (94%) – green/amber/red by verdict
Verdict badge: justified_growth | consistent | needs_review | etc.
Explanation EN (1-2 sentences)
```

Expandable "Score breakdown":

```
7 rows: dimension name + score bar + note
Bar colors: >85 green · 65-85 amber · <65 red
Weights shown as small text
```

If verdict = `justified_growth`:

```
"Explaining artifacts" chips with student notes
```

NarrativeCard.jsx

Props: `narrativeEn`, `narrativeVn`, `onRegenerate`, `loading`

Tab: English | Tiếng Việt

Paragraph text
Badge: "Generated by AI from verified data"
Button: "Refresh ↻" → POST /rebuild-narrative
Loading: 3-line skeleton
Fallback: cached if timeout

ETESTERPublic.jsx (QR destination)

Route: /etester/verify/:badgeUid (no auth)

1. Trust header: ETEST logo + green ✓ + "Verified by ETEST Vietnam"
 2. Student identity: name + program + months
 3. Requirements coverage grid: 9 items, green ✓ or gray –
 4. Verified stats table (from badge_payload)
 5. Leadership summary (if has_leadership)
 6. Trace chain summary:
"Personal Statement verified through 9 connected artifacts,
all mentor-confirmed."
 7. Narrative EN (1 paragraph)
 8. Issuer: ETEST Vietnam · etest.edu.vn
 9. Footer: issued_at · "For verification purposes only"
-

ETESTERDashboard.jsx

Route: /etester/:studentId

Layout (top to bottom):

Header: ETESTER seal + student name + "ETEST Verified Learner"
Share + Download QR buttons

Pending banner (if pending_trace_links > 0):
amber strip: "Your mentor has {N} connections to review"

Stats row: 4 metric cards
Total contributions · Mentor verified · Avg auth% · Months

Two columns:
Left: Contributor donut (Recharts)
Right: NarrativeCard

ArtifactGraph (full width)

MilestoneTimeline (full width)

QRBadge section

13. Signing Chain

Mentor sign (per artifact)
→ confirms content is authentic
→ MilestoneArtifact.mentor_signed = True

Admin sign (process verification)
→ auto-trigger after mentor sign (can be manual)

→ MilestoneArtifact.admin_signed = True

Manager issue badge (final seal)

- reviews ETESTERCore.pending_trace_links == 0
- reviews requirements_coverage
- signs JWT with ETEST private key
- ETESTERBadge created

JWT payload

```
payload = {
    "v": "1",
    "badge_uid": f"ETR-{year}-{core.id:06d}",
    "name": student.name,
    "program": student.program,
    "months": student.months_enrolled,
    "milestones": len(completed),
    "mentor_verified": core.mentor_verifications,
    "ielts": ielts_score,
    "sat": sat_score,
    "auth_avg": core.consistency_score,
    "essays_approved": essays_approved_count,
    "leadership_evidence": leadership_data.get("has_leadership"),
    "leadership_summary": leadership_data.get("ai_summary"),
    "requirements_coverage": core.requirements_coverage,
    "issued": today,
    "issuer": "ETEST Vietnam",
    "iss": "did:web:etest.edu.vn",
    "exp": expires_timestamp
}
```

Production path

- JWT RS256 (demo) → W3C VC Ed25519 (production)
- Generate Ed25519 keypair → AWS Secrets Manager
 - Publish did.json → etest.edu.vn/.well-known/did.json
 - pip install didkit → replace jwt.encode()
-

14. Build Order

- H0–H2 Data + Migration
- └─ etester_models.py (7 models, student_context_note field)
 - └─ alembic revision --autogenerate -m "etester_v4"
 - └─ alembic upgrade head
 - └─ Seed: 18–20 milestones + 9 trace links (confirmed) + forms
- H2–H4 Backend
- └─ Test 4 Claude prompts → verify JSON output
 - └─ JWT sign/verify
 - └─ GET /etester/:id → full response
 - └─ POST /contribute → trace trigger async
 - └─ GET /mentor/:id/pending + POST /verify-trace
- H4–H7 Frontend – Student view
- └─ ETESTERDashboard layout (H4)
 - └─ ArtifactGraph: solid vs dashed edges (H4–H6)

- └─ TraceLinkSuggestions post-submit screen (H5–H6)
 - └─ MilestoneTimeline + side panel (H6–H7)
- H7–H9 Frontend – Mentor view + AI
- └─ MentorDashboard: pending queue (H7–H8)
 - └─ MentorTraceReview card: confirm/reject/modify (H7–H8)
 - └─ AuthScoreCard: 7-dim breakdown (H8)
 - └─ NarrativeCard + regenerate (H8–H9)
- H9–H10 QR + Deploy
- └─ QRBadge + ETESTERPublic (requirements coverage grid)
 - └─ Railway + Vercel deploy
 - └─ Scan QR on real phone test

Seed data — 18 milestones Minh Anh

Sep 2025	ielts_mock #1	5.5
Oct 2025	mentor_session #1	focus: basics
Nov 2025	ielts_mock #2	6.0
Dec 2025	camp	Writing Camp · leadership: 8 ppl
Jan 2026	csr	Share & Care · leadership: coordinator
	5 ppl	
Feb 2026	mentor_session #5	focus: essay structure
Mar 2026	essay_draft #1	auth: 78% (seed AuthScoringResult)
Apr 2026	mentor_session #7	focus: counter-argument
May 2026	essay_draft #2	auth: 87% verdict: justified_growth
Jun 2026	ielts_mock #3	6.5 · target_achieved
Jul 2026	extracurricular members	English Club VP · leadership: 45
Aug 2026	sat_mock #1	1280
Sep 2026	award_honor	National competition · national level
Oct 2026	academic_record	HK1 GPA 9.1
Nov 2026	sat_mock #2	1320
Dec 2026	recommendation_letter	Mentor Cuong → Melbourne + RMIT
Jan 2027	mentor_session #12	approve PS
Feb 2027	essay_final	auth: 94% justified_growth ✓ approved

15. Demo Script

Minh Anh — 17t · IELTS 6.5 · SAT 1320 · 14 tháng AMP

[Student view – 2 phút]

1. Dashboard: 4 stats + requirements coverage 8/9
2. Artifact Graph: click Personal Statement Final
 - Trace chain highlights: Camp → CSR → Draft#2 → Final
 - Solid edges = mentor confirmed · student notes visible
3. Auth score: 94% justified_growth
 - "Growth explained by Writing Camp + 2 mentor sessions"
 - Show 7-dim breakdown

[Mentor view – 1 phút]

4. Switch to MentorDashboard
 - Pending queue: "3 trace links awaiting review"
 - Click 1 link: thấy AI evidence + student context note
 - Click "✓ Confirm" → edge turns solid in graph live

[QR – 30 giây]

5. Back to student view → QR Badge

- Scan bằng điện thoại
- ETESTERPublic: verified stats + trace chain summary + narrative

Close: "Khi AI viết được mọi thứ – ETESTER là bằng chứng duy nhất university tin."

Q&A

Câu hỏi	Trả lời
Ai link artifacts?	AI suggest · Student add context · Mentor confirm. Không ai làm một mình.
Auth score accurate không?	7 chiều, cần graph data. Growth lớn không bị flag nếu có artifact explain + mentor confirm.
Mentor có bị overload không?	Mentor chỉ review gợi ý của AI — không tự tạo. Review 1 link mất 10 giây với UI đơn giản.
University tin không?	ETEST 20 năm, badge signed bởi ETEST key, verify không cần account.
Scale thế nào?	Mỗi campus/center có Manager riêng. Mentor manage max_students trong model.

Appendix — What changed from v3 to v4

Change	Reason
Added Approach C linking flow (Section 4)	Core flow chưa được spec rõ
Added student_context_note to MilestoneTraceLink	Approach C requires student input field
Added pending_trace_links, pending_approvals to ETESTERCore	Mentor badge notification
Added TraceLinkSuggestions.jsx	Student sees suggestions post-submit
Added MentorDashboard.jsx	Mentor view hoàn toàn thiếu trong v3
Added MentorTraceReview.jsx	Single review card for mentor
Added MentorArtifactApprove tab in MentorDashboard	Mentor approves artifacts, not just links
Added GET /mentor/:id/pending route	Mentor fetches their queue
Added POST /student-note/:traceLinkId route	Student submits optional note
Updated Auth Scorer prompt to include student_context_note	Notes are strong evidence for D1
Updated ETESTERPublic with trace chain summary	University sees verification depth
Updated build order	Mentor UI now H7-H9, not

skipped
